

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a central framework for advancing the reasoning capabilities of large language models. Prevailing algorithms such as GRPO, however, are systematically undermined by vanishing gradients, a failure we trace to two coupled sources: the fixed clipping bounds imposed on token-level probability ratios, and the step-local standardization of rewards, which collapses to a zero advantage whenever the responses to a prompt happen to share an identical reward. Acting in concert, these two mechanisms produce ineffective gradient updates and leave a substantial fraction of generated responses unused. We introduce **Dynamic Clipping Policy Optimization (DCPO)**, which addresses both sources through two complementary and mutually reinforcing mechanisms. A dynamic clipping strategy adaptively rescales the clipping bounds as a function of token-specific prior probabilities, widening the interval for token-level exploration precisely where the old policy is least confident and the residual learning signal is most informative. A smooth advantage standardization scheme normalizes rewards over cumulative rather than step-local statistics, restoring the response-level data utilization that step-local standardization otherwise forfeits. Across four models spanning 1.5B to 14B parameters and four mathematical reasoning benchmarks, DCPO delivers the strongest overall performance, with its largest and most consistent gains concentrated on the sampling-based AIME metrics that most severely stress solution diversity. On the AIME24 benchmark with Qwen2.5-Math-7B, DCPO reaches an Avg@1 of 46.7 under greedy decoding and an Avg@32 of 38.8 under 32-sample decoding, surpassing DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9); on the AIME25 benchmark with Qwen2.5-14B, DCPO attains 23.3/19.0, likewise surpassing GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). Beyond raw accuracy, DCPO raises the nonzero-advantage ratio over GRPO by an average of 28% across the four models, attains the same update count as DAPO at roughly half its training cost, and reduces the token clipping ratio by an order of magnitude relative to both GRPO and DAPO, all while delivering stronger performance overall. Taken together, these results establish DCPO as a substantially more data-efficient approach to reinforcement learning for large language model reasoning.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central focus of contemporary research (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Within this line of work, Reinforcement Learning from Verifiable Rewards (RLVR) has established itself as a particularly effective technique: by optimizing LLMs against rule-based outcome rewards, it directly strengthens performance in domains that demand reflective, multi-step decision-making, most prominently mathematics and coding (DeepSeek-AI et al., 2024). The efficiency and stability with which RLVR converts sampled trajectories into policy improvement therefore bear directly on how far these reasoning gains can be pushed.

The GRPO (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) family of algorithms was developed to operationalize RLVR at scale. Although GRPO’s robustness has been extensively validated, subsequent analyses—most notably DAPO (Yu et al., 2025)—have exposed two critical and interrelated limitations. First, GRPO is prone to entropy collapse, in which advantage values decay toward zero in the later stages of training and cease to drive learning. Second, its indiscriminate weight clipping discards rarely sampled learning signals, curtailing the diversity of

model exploration precisely where informative updates are scarcest. Both failure modes share a common signature: a growing share of the sampled data ceases to contribute usable gradients as training proceeds.

Existing remedies address these symptoms only partially. DAPO, for instance, couples a Clip-Higher strategy, which relaxes the upper clipping bound to counteract entropy collapse, with a Dynamic Sampling strategy that filters out prompts whose responses share a uniform reward, so that every retained batch carries valid gradients. These mechanisms, however, incur two notable costs. The first is **response-level inefficiency**: dynamic sampling discards a large share of generated responses, depressing sampling efficiency and slowing convergence. The second is a **token clipping deficit**: our analysis establishes that the optimal clipping bound is token-dependent, so any single uniform upper bound is inherently suboptimal. Alternative methods that avoid outright discarding zero-gradient responses inherit closely related versions of the same limitations, which indicates that these costs are structural rather than incidental to any one design.

We introduce **Dynamic Clipping Policy Optimization (DCPO)**, which addresses both limitations through two mutually reinforcing mechanisms and a complementary loss-aggregation choice. First, we develop a **smooth advantage standardization** that normalizes rewards across cumulative rather than step-local statistics, eliminating the vanishing-gradient pathology that arises when responses sharing an identical reward are assigned a zero advantage. By interpolating the reward distribution of all cumulatively generated responses with that of the current step, this mechanism recovers otherwise discarded learning signals and thereby sustains response-level exploration deeper into training (see Section 2). Second, we propose a **dynamic-adaptive clipping** rule that adjusts the clipping bounds as a function of the old token-specific probabilities, replacing the coarse device of fixed upper and lower bounds: tokens assigned lower prior probability are granted proportionally wider bounds, opening the space for token-level exploration exactly where informative, high-entropy tokens reside (see Section 2 and Appendix A.6). Complementing these mechanisms, DCPO averages the loss over the tokens within each individual response rather than across the entire batch, a choice that preserves the relative advantage structure among responses to the same prompt and removes the dilution induced by batch-level averaging. These three design decisions constitute the paper’s central contributions, and each is isolated and validated in the component ablation of Section 5.4.

For evaluation, we construct a training corpus by merging Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of the MATH dataset (Liu et al., 2025), and we assess performance on four widely used mathematical reasoning benchmarks. Across four distinct models, with GRPO, DAPO, and—where available—GSPO as baselines, DCPO attains the strongest average performance on every model, and its advantage is most pronounced on the demanding AIME benchmarks under sampling: on AIME24-Avg@32 with 32-sample decoding, DCPO-7B scores 38.8, against 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO); on AIME25-Avg@32, DCPO-14B reaches 19.0, against 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). Beyond accuracy, DCPO raises the average nonzero-advantage ratio over GRPO by 28 percentage points, reaches a fixed update count at roughly half of DAPO’s training cost, and reduces the token clipping ratio by an order of magnitude relative to the token-level baselines, while remaining the strongest method overall.

Preliminary details, including the loss and advantage formulations of prior algorithms such as GRPO, DAPO, and GSPO, are provided in Appendix A.2.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Clipping the probability ratio is a foundational technique in reinforcement learning: it stabilizes policy updates by preventing the large, destabilizing parameter shifts that off-policy optimization would otherwise induce. This role becomes especially critical under importance sampling. Reweighting by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$ yields an unbiased estimate of $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$; the variance of that estimator, however, can be severely inflated, as quantified in Equation (1).

$$\mathrm{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \mathrm{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To curb this variance inflation, prior work (e.g., Schulman et al. (2017)) constrains the probability ratio within a fixed clipping bound governed by the single hyperparameter ϵ , as in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed clipping bound, however, constrains the model’s exploration of the diversity space, and the constraint binds most tightly in regions where the old policy assigns low probability mass: there the absolute room available to a decreasing probability is inherently narrow, throttling the capacity for meaningful policy updates. The asymmetric bounds adopted by DAPO relax this constraint only cosmetically, since the underlying probability-independent limitation persists. GSPO fares no better: by discarding entire responses whenever the sequence-level importance ratio leaves the asymmetric interval, it forfeits informative tokens while retaining high-variance ones, a combination that destabilizes training.

Governing the variance–bias trade-off therefore calls for a qualitatively different device—a probability-dependent constraint, formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term by taking its maximum with zero yields a closed-form admissible interval for $r(x)$. The result is a family of **dynamic-adaptive clipping bounds** that scale with the old probability $q(x)$ under the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

By widening the admissible interval exactly where $q(x)$ is small, this dynamic-adaptive clipping scheme restores substantial room for policy exploration in the low-probability regions that fixed bounds neglect, while leaving the interval for high-confidence tokens essentially unchanged. To guard against the pathological updates that arbitrarily large ratios could induce, we follow the dual-clipping principle of Ye et al. (2020) and cap the ratio at 10 for both positive and negative advantages, so that an overly wide bound cannot itself destabilize training. A complete derivation of these bounds from the importance-sampling framework, together with an analysis of their behavior across the probability range, is deferred to Appendix A.5 and Appendix A.6.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In prior algorithms such as GRPO, DAPO, and GSPO, the advantage $\hat{A}_{new,j}^i$ of the j -th response is obtained by standardizing only the rewards $R_{j=1,\dots,G}^i$ collected for the same prompt at the i -th step, as in Equation (11) of Appendix A.3. This step-local scheme induces two distinct failure modes. First, whenever sampling randomness renders all rewards at a step identical, the standardized advantage collapses to zero, so the prompt contributes nothing to the parameter update—even when its reasoning trajectories differ in ways that carry genuine learning signal. Second, the skewed label counts produced by high-entropy sampling drive the standardized advantage to fluctuate sharply from step to step, at times reversing sign, which destabilizes optimization.

These pathologies motivate a distributional reinterpretation of the advantage. Recent work (Yue et al., 2025) indicates that RLVR chiefly reweights the likelihood of correct samples rather than fundamentally reshaping the underlying distribution. Under this view, the rewards of the responses to a given prompt may be treated as draws from a single global distribution that persists across training, a premise that motivates a *cumulative* standardization, defined in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics are computed over all responses generated so far for the same prompt.

Neither the step-specific standardization ($\hat{A}_{new,j}^i$) nor the cumulative standardization ($\hat{A}_{total,j}^i$) is individually adequate: the former is volatile, whereas the latter is slow to track genuine policy shifts and, when the model distribution moves substantially, risks carrying stale reward statistics. We therefore interpolate between the two estimators through the pair of smoothing functions $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$, each a weighted average whose mixing weights evolve with the step index i , as defined in Equation (6). This interpolation is precisely what tempers the staleness of the cumulative statistics against the volatility of the step-local ones.

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

As training progresses, $\hat{S}A_{new,j}^i$ concentrates increasing weight on the current distribution, favoring rapid adaptation to the latest policy outputs, whereas $\hat{S}A_{total,j}^i$ shifts weight onto the cumulative distribution, treating the current step’s deviation as a fine-grained adjustment within the broader trajectory. We then take the final advantage $\hat{A}_j^i$ to be whichever of the two smoothed estimates carries the smaller absolute value, so that the sharper fluctuations of either the cumulative or the step-local standardization are damped in the interest of training stability, as defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

A key consequence follows immediately: once a prompt has entered optimization, every response it generates in subsequent steps continues to contribute, irrespective of whether its step-local advantage is zero. When identical rewards arise within a step, the affected responses still update the model through the residual term $\frac{1}{i} \hat{A}_{total,j}^i$, which preserves otherwise discarded learning signals and thereby raises data efficiency—the response-level utilization gains reported in Section 5.3 are a direct empirical manifestation of this property.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** (the loss in Equation (13), used by DAPO) equalizes the contribution of individual tokens, but in doing so it distorts the relative weighting of responses of differing lengths. Consider two responses to the same prompt: response A carries advantage 1 at length 500, while response B carries advantage 0.5 at length 1500. Under TLM, A is weighted by $\frac{500}{500+1500} = 0.25$ and B by $\frac{1500}{500+1500} = 0.75$, so that their total loss contributions become $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, respectively. The lower-advantage response B thus exerts the greater influence (0.375 versus 0.25)—precisely the inversion that length-normalized averaging is meant to avoid.

The **Sequence-Level Mean loss (SLM)** (the loss in Equation (12), used by GRPO) exhibits a complementary defect: averaging across the batch attenuates each advantage by a factor of G , and at large batch sizes this uniformly rescales the standardized advantages and thereby erases the correlation structure among them. Because standardization already encodes the relative relationships among responses to a prompt, we preserve that structure by averaging only over the $|o_i|$ tokens within each response—an operation that both retains the inter-response relationships and distributes importance uniformly across the tokens of a given response. Averaging over tokens within a single response, without a further average over the responses in the batch, defines what we term the **Only Token Mean loss (OTM)**, given in Equation (8).

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

Following DAPO, the DCPO objective omits the KL-divergence constraint, and its lower and upper clipping bounds ε_{low} and $\varepsilon_{\text{high}}$ are instantiated by the dynamic-adaptive bounds introduced in Section 2. The OTM loss, the smooth advantage standardization, and the dynamic-adaptive clipping thus combine within a single objective, each addressing a distinct source of wasted gradient signal.

4 EXPERIMENTAL SETUP

Models and Datasets To probe generality across model scales and pretraining regimes within the Qwen2.5 family, we evaluate on four models spanning both math-specialized and general-purpose bases: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). Training uses a merged corpus of approximately 25k mathematical problems that combines the DAPO-Math-17K dataset (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). Every problem is rendered with the Qwen-Math template (see Appendix A.8) for both training and evaluation, so that the input format remains consistent throughout and identical across all compared methods.

Evaluation We evaluate on four widely used mathematical reasoning benchmarks that together span a broad difficulty spectrum: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, whose larger size (500 problems) already yields a stable estimate, we report Avg@1. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) benchmarks, we report both Avg@1 and Avg@32, so that the evaluation captures greedy accuracy and sampling-time robustness jointly. To keep the comparison strictly controlled, all methods are evaluated under identical sampling temperature, prompt template, and answer-extraction rules.

- **Avg@1**: accuracy under greedy decoding, which measures the quality of the model’s single most-likely prediction.
- **Avg@32**: the mean accuracy over 32 responses sampled per problem at temperature 1.0 and top_p 1.0, which characterizes the robustness and stability of the trained policy distribution under stochastic decoding.

Baseline Settings We adopt GRPO and DAPO as baselines throughout, and augment the Qwen2.5-Math-7B and Qwen2.5-14B settings with GSPO as an additional, stronger baseline. To keep the comparison fair under a common compute budget, all methods share the same training corpus, template, generation and mini-batch sizes, learning rate, and update count; the only settings that differ are the method-specific clipping thresholds ϵ_* , which are set to each method’s recommended values (Appendix A.9). Because DAPO’s dynamic sampling regenerates responses to refill filtered batches, matching the update count entails a larger generation budget for DAPO, an asymmetry we quantify directly in Section 5.3. The remaining configuration—the codebase and the shared training hyperparameters—is detailed in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1 reports the performance of GRPO, DAPO, GSPO, and our proposed DCPO on the four mathematical reasoning benchmarks—MATH500, AMC23, AIME24, and AIME25—under both greedy decoding (Avg@1) and 32-sample decoding (Avg@32), so that robustness and generalization are assessed jointly rather than in isolation. Across every model scale, from 1.5B to 14B and across both math-specialized and general-purpose Qwen2.5 bases, DCPO attains the best average score, and its per-benchmark results are the best or closely competitive throughout. The consistency of this advantage across four models of differing capacity and pretraining regime indicates that DCPO functions as a broadly applicable policy optimization technique for RLVR-based mathematical reasoning, rather than a method tuned to any single model or benchmark. We report single-run results per configuration under a fixed compute budget; where methods differ by a point or less on an individual benchmark, we read the outcome as a tie rather than a decisive gap, and rest the central claims on the averaged and sampling-based metrics, where the margins are largest and most stable.

Overall Superiority of DCPO Reading the averaged Avg@1/Avg@32 columns, DCPO leads on all four models. On Qwen2.5-Math-1.5B-Instruct it attains the highest averages (47.2/32.8), exceeding GRPO by (+1.2/+0.8) and DAPO by (+0.7/+0.4). On Qwen2.5-3B it reaches (37.6/22.7), improving on GRPO by (+1.3/+1.7) and leading DAPO on average, combining a clear Avg@1 gain (+1.0) with a marginal Avg@32 shortfall (-0.4) that we regard as a statistical tie. On Qwen2.5-Math-7B it attains (57.1/45.2), surpassing GRPO by (+4.0/+3.6), DAPO by (+3.3/+2.8), and GSPO by (+1.9/+1.9). On the largest model, Qwen2.5-14B, it delivers (53.2/39.0), improving substantially

Table 1: Performance across the four benchmarks for each model, under a fixed compute budget with a single run per configuration. Avg@1 denotes greedy decoding and Avg@32 the mean over 32 samples at temperature 1.0. Boldface marks the best value in each column; each Δ row reports DCPO’s difference from the indicated baseline (GRPO, DAPO, or GSPO), with gains in red. The rightmost **Average** column, over which DCPO leads on every model, provides the most stable summary comparison.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

over GRPO (+6.5/+7.7) and outperforming both DAPO (+1.3/+0.1) and GSPO (+4.2/+5.5) on both metrics. The gains are not uniform at the level of every individual benchmark—DCPO trails the strongest baseline on a handful of columns, such as MATH500 against GSPO at 7B and AMC23-Avg@32 against DAPO at 14B—but the averaged view is unambiguous: the advantage is consistent across model capacities and holds under both greedy and stochastic decoding, evidencing a policy that aligns more faithfully with the reward signal irrespective of the decoding regime.

Robustness on AIME under Sampling (Avg@32) The most pronounced pattern emerges on AIME24 and AIME25, the high-difficulty competition benchmarks that most severely stress a model’s reasoning ability. Under Avg@32—which measures whether the model can produce correct responses under high-entropy sampling—DCPO’s robustness advantage widens markedly at the 7B and 14B scales. On AIME24-Avg@32, DCPO-7B achieves 38.8, outperforming GRPO (32.1), DAPO (31.6), and GSPO (34.9), a margin of +3.9 over the strongest baseline; on AIME25-Avg@32, DCPO-14B reaches 19.0, exceeding GRPO (10.5), DAPO (15.3), and GSPO (9.9), a margin of +3.7. Because these gains surface specifically under sampling rather than under greedy decoding, they

indicate that DCPO improves not merely average-case accuracy but the diversity and reliability of correct generations—the property that matters most whenever temperature-driven variability or multi-sample decoding is in play. Improving reward fidelity and exploration jointly under high-entropy decoding is exactly the behavior sought in deployment settings that draw multiple candidate outputs per query, and it aligns directly with the response-utilization and clipping mechanisms analyzed in the following subsections.

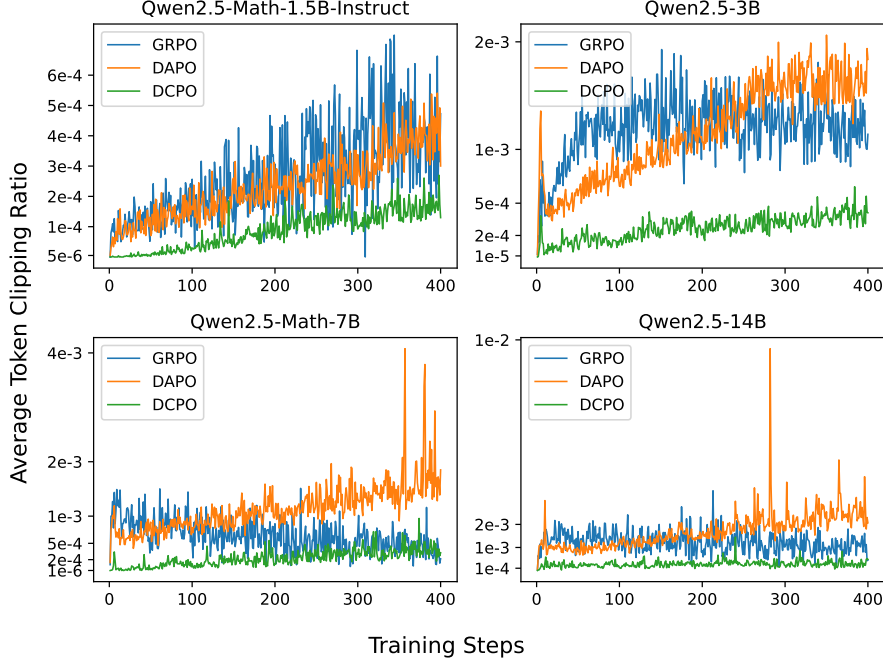


Figure 1: Token Clipping Ratio (*TCR*) over training for each method across the four models. Lower, flatter curves indicate that fewer tokens are excluded from policy updates. DCPO (our method) stays an order of magnitude below GRPO and DAPO throughout, and does so without the intermittent spikes that mark the baseline curves.

5.2 THE TOKEN CLIPPING RATIO (*TCR*)

We define the **Token Clipping Ratio** (*TCR*) as the proportion of tokens excluded from policy updates during back-propagation as a result of clipping; it directly quantifies how much of the generated signal each method forgoes. As Figure 1 shows, the *TCR* diverges sharply across the reinforcement learning methods, exposing large differences in how efficiently each converts generated tokens into gradient signal.

$$TCR = Average \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } micro_m}{\text{Total number of tokens in } micro_m} \right) \quad (9)$$

where $micro_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

Under GRPO, the *TCR* trend is model-size dependent: for the smaller models (Qwen2.5-Math-1.5B and Qwen2.5-3B) it rises with training steps, whereas for the larger models (Qwen2.5-Math-7B and Qwen2.5-14B) it gradually declines. DAPO, by contrast, exhibits a monotonically increasing *TCR* across all models, so that as training proceeds an ever-larger share of policy updates rests on partial or truncated responses. Both GRPO and DAPO, moreover, display pronounced fluctuations and intermittent spikes in the *TCR*—a source of optimization instability that can precipitate entropy collapse. GSPO is more wasteful still: because it clips at the sequence level, its average *TCR* exceeds 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B, far above the token-level clipping methods, and it discards a correspondingly larger volume of informative tokens.

DCPO, in sharp contrast, sustains a stable and efficient training regime. Independently of model size or training stage, its *TCR* remains nearly constant and an order of magnitude below those of GRPO and DAPO. This lower and steadier *TCR* means that DCPO discards far fewer tokens, so that a greater fraction of each complete response informs the policy update while more room is reserved for exploration over rare tokens. The net effect is a substantial sample-efficiency advantage over the baselines, obtained while the space for outlier variation stays tightly controlled.

The token statistics of Qwen2.5-Math-7B make the mechanism concrete: the token distribution rapidly concentrates on high-confidence tokens, with roughly 95% of generated tokens satisfying $q(x) > 0.9$ after about 60 steps, rising to roughly 97% after about 100 steps and increasing thereafter. As illustrated in Figure 4d, once the old probability satisfies $q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$, the new probability $p(x)$ enters the update within the interval $[\frac{1}{1+\epsilon}, 1]$ and is retained under any clipping scheme. The overwhelming majority of tokens therefore lie in a never-clipped region, which accounts directly for the low overall *TCR* observed in practice. At the same time, Figure 1 shows the *TCR* climbing in the later phases of most training runs, signaling that an increasing fraction of low-probability tokens is being discarded—precisely the tokens whose exclusion most restricts the exploration of rare-token diversity.

This distinction matters because prior work (Wang et al., 2025) reports that **high-entropy tokens**, which coincide with the low-probability tokens, are among the principal drivers of emergent reasoning ability. As entropy and probability are inversely related, rarer tokens carry proportionally more information; widening their admissible update range enlarges the model’s effective exploration space and plausibly facilitates the acquisition of diverse reasoning behaviors.

These empirical patterns, together with the preceding observation, motivate our **dynamic-adaptive clipping** scheme, which relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the tight, safe interval for high-confidence tokens. By adaptively expanding the update interval over the high-entropy tail of the distribution, the method restores the model’s capacity to learn from rare, informative tokens, an effect consistent with the reasoning gains reported in Section 5.1.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a response contributes to the policy update only when it carries a nonzero gradient. We quantify how effectively each method exploits its generated data through the **Response Utilization Ratio** (*RUR*), defined as the percentage of generated responses that possess a nonzero advantage and therefore participate in the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

As Table 2 shows, the *RUR* under GRPO is low—averaging 43.8% across the four models—so that more than half of all generated responses never inform the policy, a marked inefficiency in exploiting the sampled data. DCPO reverses this pattern: after the first epoch it sustains an average *RUR* of roughly 70%, an absolute increase of 28 percentage points and a relative improvement of 64% over GRPO, and it does so consistently across every model. This is the response-level counterpart to the smooth advantage standardization of Section 2: the residual cumulative term keeps prompts contributing gradients that step-local standardization would zero out.

Table 2: Average *RUR* over 400 training steps, by model and method. Higher values indicate that a larger share of generated responses contributes to policy updates. DCPO (our method) utilizes roughly 70% of responses on every model, well above GRPO (43.8% average) and GSPO (45.6% average). A dash (-) marks configurations for which GSPO was not evaluated; DAPO is excluded because its dynamic sampling discards uniform-reward prompts before training, which makes a direct *RUR* comparison ill-defined.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 makes the temporal dynamics explicit: under GRPO the *RUR* degrades substantially over training, irrespective of model size or type, falling sharply from above 90% to below 50%, and

on Qwen2.5-Math-7B collapsing to as low as 30% after roughly 200 steps. This decay is the direct signature of GRPO’s step-local advantage standardization, which drives a growing fraction of gradients to zero and renders exploration progressively inefficient in the later stages of training. DCPO, by employing smooth advantage standardization, avoids this collapse: after the first epoch its *RUR* quickly stabilizes near 70% and continues to rise with further training.

The shape of DCPO’s *RUR* trajectory—above 90% in the opening steps, then settling around 70% from the second epoch onward without reaching 100%—is itself informative, and we analyze its two regimes below to make the role of the SAS mechanism explicit.

The elevated early-stage *RUR*, exceeding 90%, stems chiefly from the early model’s frequent violations of the required output format, which draw negative rewards and inflate the ratio; because models learn these fixed answer patterns quickly, subsequent training can redirect emphasis from surface formatting toward genuine reasoning capability. The steady-state value near 70% likewise admits a principled explanation, resting on two sources of persistently zero advantage. (i) Nearly half of the problems are easy enough that the model solves them correctly and consistently; excluding these from policy updates is warranted, since the model’s core competence on them requires no further refinement. (ii) The remainder are answered consistently incorrectly, for one of two reasons: some problems simply exceed the current model’s reach even given a correct ground-truth label, while others reflect label noise, in that the DAPO-Math-17K dataset contains items whose labels were coerced into mislabeled integers and thus introduce spurious “incorrect” targets. In both cases, discarding zero-advantage responses acts as an implicit filter that shields the model from updates driven by unsolvable or mislabeled data—so that the 70% steady state reflects appropriate selectivity rather than residual inefficiency.

Because DAPO discards responses with identical labels before they enter training—unlike GRPO and DCPO—a direct *RUR* comparison is ill-defined, so we instead quantify DAPO’s efficiency through the resources it consumes to reach a fixed number of update steps: the count of generated responses and the wall-clock training time. To offset the low sampling efficiency induced by GRPO’s low *RUR*, we set DAPO’s generation batch size to 1–4 times the training batch size and adjust it dynamically. The outcome is unambiguous: across the four models, DAPO requires 3 to 5 times as many generated responses as DCPO to complete the same number of parameter updates, at least doubling the training GPU hours while yielding no performance gain, and its data utilization deteriorates further with scale—on Qwen2.5-14B, the fraction of responses actually used falls below 30% after roughly 300 steps. Dynamic sampling thus couples low data utilization and low training efficiency with a forfeited opportunity to explore model diversity. This comparison also directly addresses the fixed-sample regime: at a matched generation budget, DAPO’s filtering leaves far fewer usable updates, so DCPO’s advantage does not rest on consuming more samples.

In sum, DCPO advances beyond both GRPO and DAPO: by markedly raising response utilization and training efficiency, it converts the full pool of sampled responses into a larger exploration space for model diversity, and translates that efficiency into stronger performance.

5.4 ABLATION STUDIES

To isolate the contribution of each DCPO component, we conduct a controlled ablation on Qwen2.5-Math-7B using **Avg@32**, the metric most sensitive to the robustness and stability of the learned policy distribution. Each variant alters exactly one component of the baseline GRPO framework while holding all other settings fixed and removing the KL-divergence term, which aligns the setup with DAPO, GSPO, and the full DCPO. The corresponding learning curves over steps 20 to 400 appear in Figure 3; because the study is confined to a single model, we read it as a mechanism-level diagnostic that complements, rather than substitutes for, the cross-scale results of Table 1.

The three single-component variants each improve reasoning ability throughout training, surpassing GRPO and matching or exceeding DAPO and GSPO. Every modification thus confers a distinct benefit, and their combination compounds these benefits into a substantial aggregate gain over the baselines. The separation widens in the later stages of training: whereas the baselines plateau, DCPO continues to trend upward, both at the component level and in aggregate. We detail each contribution in turn.

(i) **Only-Token-Mean loss (OTM)**: aggregating the loss within each response rather than across the whole batch already outperforms the baselines, which suffer from lower data utilization and more

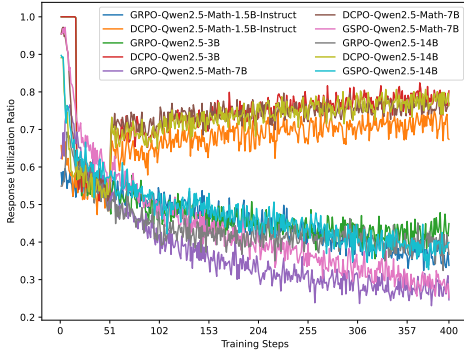


Figure 2: Evolution of the *RUR* over training. GRPO decays from above 90% to below 50%, whereas DCPO (our method) stabilizes near 70% after the first epoch and rises thereafter.

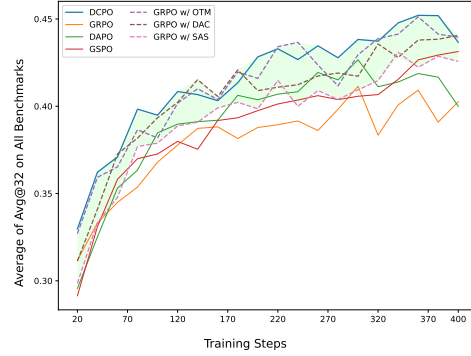


Figure 3: Component ablation on Qwen2.5-Math-7B, measured by average Avg@32 over steps 20–400. Each DCPO component (OTM, SAS, DAC) is added individually to the GRPO baseline, and the full DCPO combines all three; every single-component variant surpasses GRPO, and their combination is the strongest and most stable curve.

volatile performance. (ii) **Smoothed Advantage Standardization (SAS)**: replacing the step-local advantage with the cumulative smoothed advantage markedly improves on GRPO and approaches DAPO, corroborating that aggregating reward statistics across steps stabilizes optimization and lifts final performance. (iii) **Dynamic Adaptive Clipping (DAC)**: substituting GRPO’s fixed clipping boundary with the probability-adaptive boundary yields the largest single-component gain—DAC alone surpasses GRPO, DAPO, and GSPO, and is more stable than OTM in isolation. (iv) **OTM + SAS + DAC (full DCPO)**: integrating all three mechanisms delivers the best and most stable curve, clearly outperforming GRPO, DAPO, and GSPO. Because each component targets a distinct source of wasted gradient signal—batch-level dilution, zero step-local advantages, and probability-independent clipping, respectively—their benefits are complementary rather than redundant, and the combination improves performance and robustness jointly.

Overall, the ablation indicates that every component of DCPO contributes positively on this model and that their combination produces substantial cumulative gains, supporting the proposed mechanisms as effective means of improving data efficiency and stability in reinforcement learning for LLM reasoning.

6 CONCLUSION

This work targets two coupled limitations of existing policy optimization methods: the restricted token-level exploration imposed by fixed clipping in low-probability regions, and the depressed sample efficiency imposed by step-local reward standardization. We address both with **Dynamic Clipping Policy Optimization (DCPO)**, a reinforcement learning method that strengthens the reasoning capabilities of large language models through two complementary innovations: a **dynamically adaptive token-level clipping** mechanism that rescales the bounds according to the old-policy probabilities, enabling efficient exploration of rare, low-probability tokens; and a **cumulative-smooth advantage standardization** that aggregates the reward distribution across optimization steps, damping the reward randomness introduced by high-entropy sampling. The Only-Token-Mean loss couples the two by preserving the inter-response advantage structure that batch-level averaging would otherwise dilute. Experiments on four mathematical reasoning benchmarks across four Qwen2.5 models substantiate DCPO’s effectiveness: the DCPO-7B variant, for instance, reaches an Avg@32 of 38.8 on the demanding AIME24 benchmark, outperforming GRPO (32.1) and DAPO (31.6). Beyond accuracy, DCPO raises the response utilization ratio by 28 percentage points over GRPO, reaches a fixed update count at roughly half of DAPO’s training cost, and reduces the token clipping ratio by an order of magnitude—concrete evidence of substantially improved data utilization and diversity exploration in RLVR. Because these mechanisms are formulated over token probabilities and verifiable rewards rather than any math-specific structure, a natural and well-motivated direction

for future work is to evaluate DCPO on other verifiable-reward domains, such as code generation and tool use, and to characterize its sensitivity to the clipping hyperparameters across settings.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In prior algorithms, including GRPO and DAPO, the advantage $\hat{A}_{j,t}^i$ of token t in response j is computed by standardizing the reward R_j^i against the mean μ^i and the standard deviation σ^i of the rewards of the G responses generated at the i -th step, as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

Whenever the rewards for a given prompt are identical, the resulting zero advantage prevents those responses from contributing to the update, so the corresponding generated data is wasted—the precise failure mode that the smooth advantage standardization of Equation (5) is designed to remove.

A.3.1 GRPO

Shao et al. (2024) introduced **Group Relative Policy Optimization (GRPO)**, which samples G responses per query, assigns rewards R through a rule-based reward function, and estimates token-level advantages as in Equation (11). The parameters are then updated with Equation (12), the **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages over all tokens within a response and then over the G responses in the batch; the $\mathbb{D}_{\text{KL}}$ penalty constrains the policy against excessive deviation from a reference policy (e.g., a pre-trained model); and the fixed clipping threshold ε bounds the change in token probability ratios, precluding overly large updates. Widely adopted as it is, this two-stage averaging inherently dilutes the importance of individual responses—the specific effect that motivates the OTM loss of Equation (8).

A.3.2 DAPO

Building on this, Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)**, motivated by the observation that under GRPO longer sequences exert a disproportionate influence on the gradient. DAPO normalizes by the token count across the entire batch, yielding the **Token Level Mean loss (TLM)** of Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. This constraint discards every prompt whose responses all share an identical reward and regenerates responses to restore the batch size; while it guarantees valid gradients, it does so at the cost of severe data inefficiency, as the response-utilization analysis of Section 5.3 quantifies.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping injects high-variance noise into the gradient estimate—noise that accumulates over long sequences and is further amplified by the clipping operation itself. GSPO mitigates this by replacing token-wise clipping with sequence-wise clipping, yielding a more stable training signal at the sequence level.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_\theta(o_{j,t}|q)}{\pi_{\theta_{old}}(o_{j,t}|q)} \right) \right)$$

This stability, however, comes at a cost: GSPO discards more than 10% of responses that carry non-zero advantages, wasting substantial usable data. GRPO, conversely, suppresses responses with high sequence-level variance yet retains many with high token-level entropy, and this residual leaves training prone to instability. DCPO instead operates at the token level with a probability-dependent bound, retaining informative low-probability tokens rather than discarding whole responses.

A.4 REWARD CALCULATION

To supply a verifiable and interpretable supervision signal, we employ a rule-based reward model that scores each generated response jointly on answer correctness and format compliance, as specified in Equation (15). Concretely, a response earns +1 when both its format and its answer are correct, 0 when the format is correct but the answer is wrong, and −1 when the format itself is invalid. This three-level scheme separates the format-learning phase from the reasoning-learning phase, a separation that the early-stage *RUR* dynamics of Section 5.3 make visible.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response within the group generated for a given problem at the i -th training step, and the reward is determined by the response's format validity and answer correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Clipping the probability ratio is a crucial device for stabilizing policy optimization in reinforcement learning, since it prevents the large, destabilizing parameter updates that off-policy learning can otherwise induce. Its role becomes central when importance sampling is used to estimate expectations under a new policy from samples generated by an old one, and it is from this setting that we derive the dynamic-adaptive bounds.

The expected value of a function $f(x)$ under the new probability $p(x)$ can be re-expressed as an expectation under the old probability $q(x)$ through the importance-sampling weight, as in equation 16:

$$\int f(x)p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, its variance can be severely inflated—a well-known hazard of importance sampling. The true variance of $f(x)$ under the new policy appears in Equation (17), whereas the variance of the importance-sampled estimator under the old policy appears in Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} [f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x) \frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The gap between the importance-sampled variance and the true variance quantifies this inflation exactly, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To contain this difference, many prior methods—including PPO Schulman et al. (2017) and subsequent RLVR approaches (DeepSeek-AI et al., 2024; Liu et al., 2025)—impose a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, inherently caps the policy's capacity for exploration, and the effect is most damaging where the old policy assigns low probability mass: there the absolute room afforded to a decreasing probability is severely limited, throttling meaningful updates. Asymmetric bounds of the kind DAPO adopts soften but do not remove this defect, since the underlying limitation of a non-adaptive constraint persists.

Recent studies, notably Yue et al. (2025), further indicate that the reasoning paths exploited by RLVR models already reside within the sampling distribution of the base model—a claim corroborated by the minimal parameter shift observed during off-policy batch updates and reflected in the small fraction of clipped parameters. Motivated by this stability, we replace the conventional fixed clipping of

Equation (20) with a dynamic-adaptive mechanism that folds the probability itself into the constraint on $r(x)$, as in Equation (21).

$$|(\frac{p(x)}{q(x)} - 1)p(x)| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality and expanding yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For the resulting clipping ratio to be mathematically valid and meaningful, it must respect the fundamental constraints on probability distributions and their ratios. As stated in Equation (23), the probabilities $p(x)$ and $q(x)$ lie in $[0, 1]$, from which it follows that both the ratio $r(x)$ and the clipping hyperparameter ϵ are non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To guarantee both the validity and the practical applicability of the clipping bound derived from Equation (22), we state the necessary conditions on its endpoints in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with Equation (24), the probability ratio $r(x)$ must satisfy the following conditions:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

To fix the clipping bound, we restrict attention to the overlapping region centered on the ratio value 1, which minimizes the discrepancy between the two conditions while preserving consistency with GRPO's conventional fixed bound; the resulting validity conditions are:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we keep the radicand non-negative by taking its maximum with zero, i.e., $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This yields the **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower clipping bounds in Inequality 27, we assign distinct threshold hyperparameters ϵ_{high} and ϵ_{low} , chosen so that the dynamic bound coincides with GRPO's fixed bound at specific reference points; the calibration and its consequences are analyzed in Appendix A.6.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To align the dynamic-adaptive clipping bounds with the fixed symmetric bounds at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we obtain the specific values of ϵ_{low} and ϵ_{high} by solving the system in Equation (28).

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \\ q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases} \quad (28)$$

Under the standard practice (e.g., GRPO) of setting the fixed threshold $\epsilon = 0.2$, this calibration yields clipping hyperparameters of approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Because these values are derived from the alignment condition rather than tuned per model or dataset, the same setting transfers unchanged across all four models in our experiments. Crucially, although $\epsilon_{low} \neq \epsilon_{high}$, this asymmetry serves a purpose entirely distinct from the asymmetric fixed clipping of DAPO: here it enforces alignment with GRPO’s bound at the reference points rather than merely widening the upper bound.

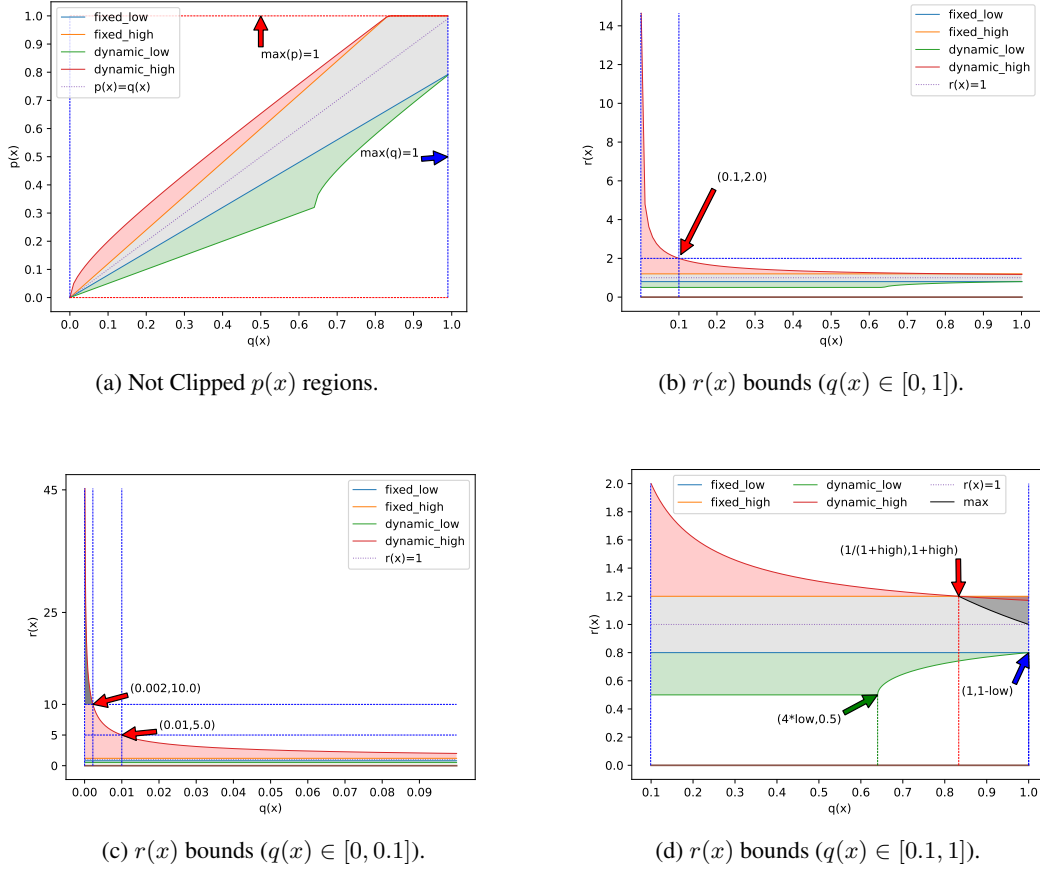


Figure 4: Comparison of clipping bounds across the probability range. The curves contrast fixed clipping ($\epsilon = 0.2$) with dynamic-adaptive clipping ($\epsilon_{low} = 0.16, \epsilon_{high} = 0.2$), showing how the adaptive bounds widen the admissible interval for low-probability (high-entropy) tokens while coinciding with the fixed bounds for high-confidence tokens. This probability-dependent widening is the mechanism behind the reduced token clipping ratio of Figure 1.

Dynamic versus fixed clipping: Figure 4 contrasts the behavior of GRPO’s fixed clipping scheme ($[1 - \varepsilon, 1 + \varepsilon]$) with our probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed bound supplies a constant relative clipping width of 2ε , so that as $q(x)$ falls the absolute exploration space it permits shrinks linearly (Figure 4a). The relative width of the dynamic scheme, by contrast, scales inversely with $q(x)$, so that the exploration space remains approximately constant across the entire probability range.

Effect of decreasing $q(x)$: Because the fixed admissible interval $[1 - \varepsilon, 1 + \varepsilon]$ is independent of $q(x)$, shrinking $q(x)$ does nothing to improve the model’s exploration of low-probability tokens. The dynamic-adaptive bound instead expands rapidly as $q(x)$ decreases (Figure 4b), granting rare tokens a larger relative update space—precisely the behavior that efficient exploration requires.

Behavior for extremely low probabilities: Figure 4c depicts the clipping interval for $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound on $r(x)$ grows in inverse proportion to the square root of $q(x)$, whereas the fixed bound stays constant. Following the dual-clipping strategy of Ye et al. (2020), we impose a hard ceiling $r_{\max} = 10$ to forestall the attendant instabilities (vanishing or exploding gradients). The gray region in Figure 4c, corresponding to approximately $q(x) \leq 2 \times 10^{-3}$, marks where the adaptive upper limit saturates at $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$ the dynamic-adaptive scheme yields $dynamic_{low} = 0.5$, while the fixed scheme holds the constant $fixed_{low} = 1 - \varepsilon = 0.8$; as $q(x)$ rises past this threshold, the adaptive lower bound contracts smoothly toward the fixed value. Symmetrically, the adaptive upper bound exceeds the fixed one whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$ and merges with it for larger $q(x)$. The adaptive interval $[dynamic_{low}(q(x), \epsilon_{low}), dynamic_{high}(q(x), \epsilon_{high})]$ therefore widens the admissible region in a probability-dependent manner for rare tokens while retaining the tightness of the fixed interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) comprises both a positive- and a negative-advantage term. Under the minimum-value operation, a positive advantage clips $r(x)$ to $[0, dynamic_{high}(q(x), \epsilon_{high})]$, whereas a negative advantage clips it to $[dynamic_{low}(q(x), \epsilon_{low}), \infty)$. Because an excessively large $r(x)$ would destabilize training, we cap the ratio at $r_{\max} = 10$ for both advantage signs; this safeguard, again following Ye et al. (2020), ensures that the adaptive limits never produce excessively large importance weights.

In short, for both positive and negative advantages, dynamic-adaptive clipping affords the model substantially more room to explore diversity when rare tokens carry low probability than fixed clipping allows, while remaining bounded against instability.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i index the iteration at which prompts are used to generate responses and update the model parameters. We now define the statistics used for reward normalization in the smooth advantage standardization.

The mean rewards of the responses are given by the group of functions in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total} = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

Correspondingly, the variances of the rewards are given by the group of functions in Equation (30):

$$\begin{cases}
\sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\
\sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{old}^i)^2} \\
\sigma_{total}^i = \sqrt{\frac{1}{G * i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^i)^2} \\
= \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)}
\end{cases} \quad (30)$$

To reduce resource usage, we compute σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29) through the equivalent recursions in terms of μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i , rather than by directly aggregating the original rewards. This recursive form makes the cumulative standardization inexpensive, adding only constant-memory bookkeeping per prompt rather than storing the full reward history.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments in this paper adopt the following template, derived from the Qwen-Math template (Yang et al., 2024b) and applied identically across every method and benchmark.

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

We configure each method’s clipping thresholds at its recommended values: a symmetric $\epsilon = 0.2$ for GRPO; lower/upper thresholds (0.2, 0.28) for DAPO, together with an additional 512-token soft-punish cache and a maximum probability ratio of 10 for negative advantages; lower/upper thresholds ($3e-4$, $4e-4$) for GSPO; and low/high thresholds (0.16, 0.2) for DCPO. As derived in Appendix A.6, the DCPO thresholds are not tuned but are fixed by the requirement that the dynamic bound preserve the same clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1 + \epsilon)$ and $(q(x), r(x)) = (1, 1 - \epsilon)$ as GRPO (Figure 4d); the identical setting is used for all four models. Unlike DAPO, DCPO caps the maximum probability ratio at 10 for both positive and negative advantages.

Each run comprises 400 training steps with a generation batch size of 512 and a mini-batch size of 32 for parameter updates. Using temperature 1 and top_p 1, we sample $G=16$ responses per problem, with the prompt length capped at 1,024 tokens and the response length at 3,072 tokens—substantially below the 20k maximum used in the original DAPO. These settings are held fixed across all compared methods so that differences in outcome are attributable to the algorithms rather than to divergent budgets.

Our training codebase is adapted from Verl (Sheng et al., 2024) and runs on $32 \times H20$ GPUs. Unless otherwise stated, hyperparameters follow those of DAPO, so that the baselines are evaluated under their own recommended configuration. For loss computation we retain the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulations of GRPO and DAPO, rather than Verl’s approximate weighting, thereby ensuring consistency across microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

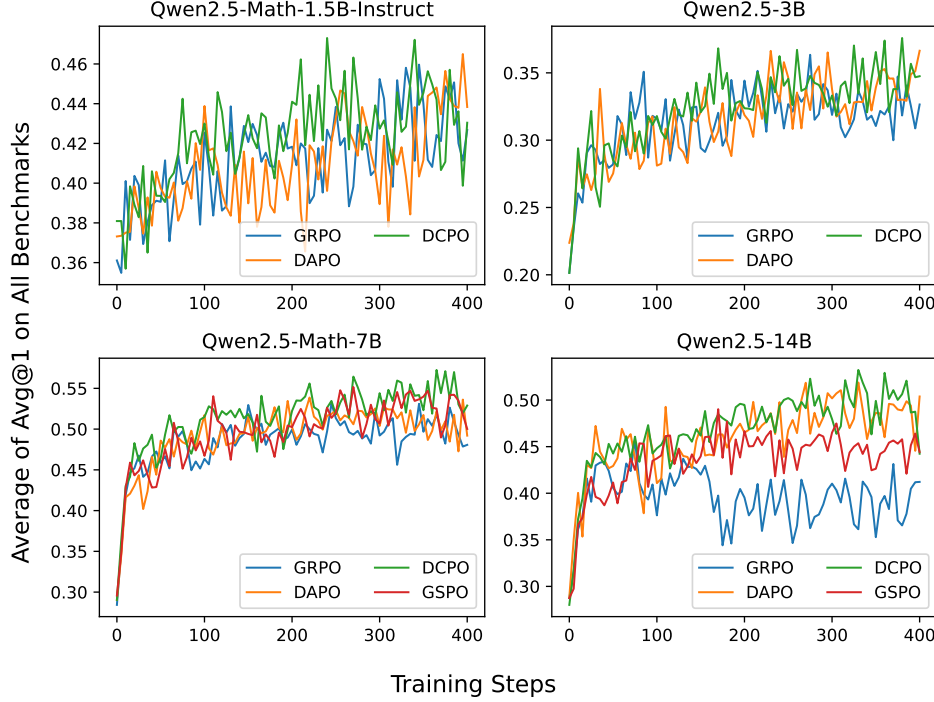


Figure 5: Avg@1 performance over training, averaged across the four benchmarks, for each model and method. DCPO (our method) both leads and trains more smoothly than the baselines across all model scales.

Avg@1 Trend Figure 5 reports the average Avg@1 metric, obtained by averaging the greedy-decoding results over MATH500, AMC23, AIME24, and AIME25, which reflects the model’s best-prediction performance. DCPO outperforms GRPO and DAPO on all four models and outperforms GSPO on the two models where GSPO was evaluated. Its trajectory is also markedly more stable than the baselines’, exhibiting neither the pronounced performance drop of GRPO nor the large fluctuations of DAPO. Taken together, these trends indicate that DCPO elicits stronger reasoning capability during training while leaving greater headroom for further improvement.

Avg@32 Trend Figure 6 compares the Avg@32 trajectories of DCPO, GRPO, DAPO, and—where available—GSPO across model sizes on AMC23, AIME24, and AIME25, and the contrast is pronounced. GRPO fluctuates sharply and occasionally collapses—on Qwen2.5-14B its performance drops abruptly after roughly 170 steps—whereas DCPO rises consistently and with minimal fluctuation across every model from 1.5B to 14B. DCPO also achieves the fastest early growth and sustains its upward trend thereafter, and relative to DAPO it attains superior performance while consuming roughly half the training time and resources. Together, these results indicate that dynamic-adaptive clipping and smooth advantage standardization jointly improve exploration at both the response level and the label level of diversity, and do so robustly across model scales.

Entropy Trend Figure 7 shows that GRPO undergoes a sharp entropy decay after an initial warm-up phase and quickly settles into a flat, low-entropy regime. This entropy collapse signals a rapid loss of policy randomness that curtails the diversity of sampled responses and thereby constrains subsequent optimization.

The entropy dynamics of DAPO and DCPO are model-dependent. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories, but on the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B) they diverge substantially. DAPO sustains the highest entropy, rising clearly but with large fluctuations. On Qwen2.5-Math-7B, GSPO’s

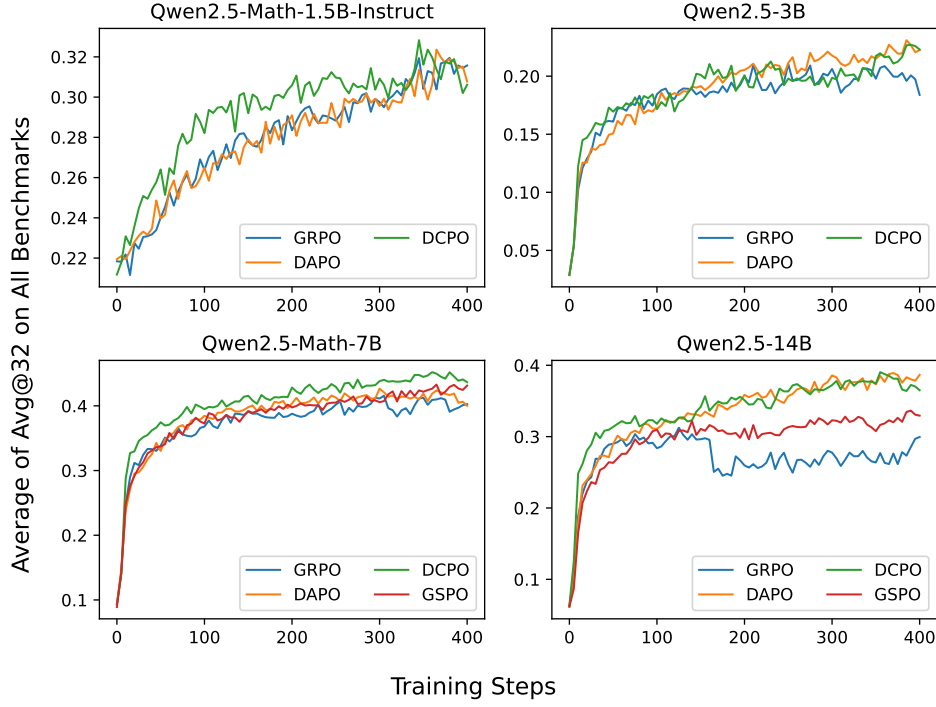


Figure 6: Avg@32 performance over training on the AMC23, AIME24, and AIME25 benchmarks, for each model and method. DCPO (our method) rises fastest early and sustains the trend, whereas GRPO fluctuates and occasionally collapses.

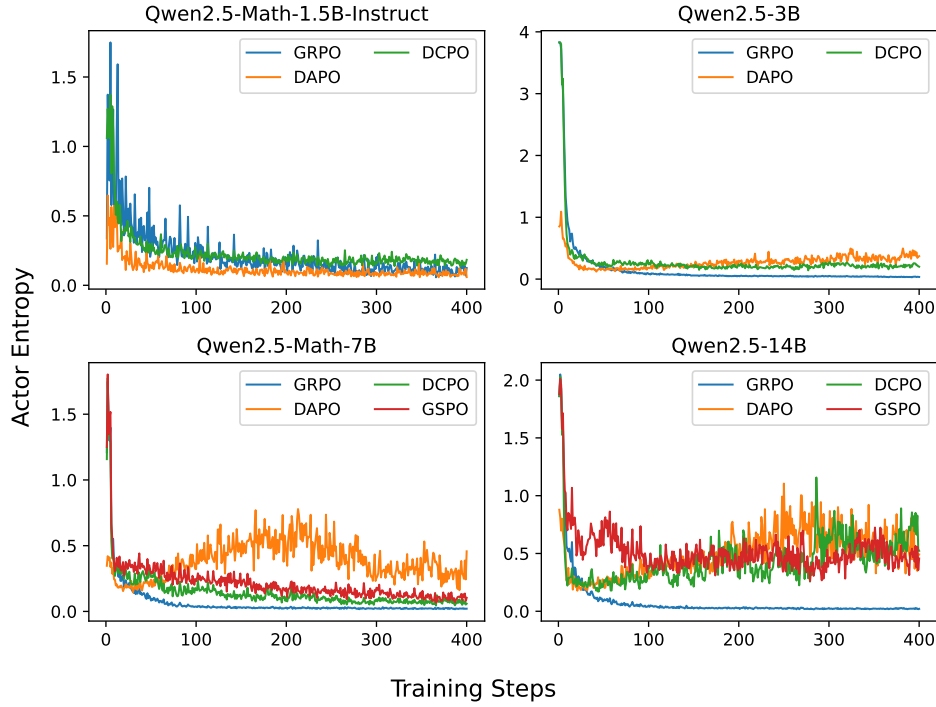


Figure 7: Policy entropy over training for each model and method. DCPO (our method) holds entropy within a moderate band, avoiding both GRPO’s collapse and DAPO’s high-variance growth.

entropy tracks that of DCPO and their average performance is comparable, whereas on Qwen2.5-14B GSPO’s entropy trends downward with heavy fluctuation and its performance suffers accordingly. DCPO, by contrast, settles at a moderate entropy: its curve is conspicuously smoother and remains confined to a bounded band between GRPO’s low-entropy region and DAPO’s high-variance region.

These observations are consistent with a U-shaped relationship between entropy and performance: too little entropy starves the learner of exploratory signal and induces premature convergence, whereas too much—or too unstable—entropy undermines training stability. By holding entropy within a moderate, well-balanced band, DCPO secures both convergence and sufficient exploration, and this advantage becomes more pronounced with model capacity. This entropy-regulation behavior offers a plausible account of DCPO’s stronger reasoning performance and its robust policy learning across model scales.